

Exploring IoT and Machine Learning Integration for Soil Nutrients Monitoring and Crop Recommendation: A Survey

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Abstract— The integration of machine learning (ML) and Internet of Things (IoT) technologies has a scope of improvement in precision farming techniques and revolutionise the agriculture sector. This research paper examines the integration of IoT and ML in soil nutrient monitoring and crop recommendation systems. The aim is to provide a thorough overview of the most recent developments, applications, challenges, and potential future directions. IoT sensors collect and give immediate soil health information, while ML algorithms analyses these data to suggest nutrient deficits and recommend the most suitable crops. This research work evaluates advancements in IoT and Machine Learning applications, emphasising their benefits for precision agriculture. This integration aims to improve agricultural production and sustainability via data driven decision making and resource optimisation. At the end research gaps are discussed, along with future research directions.

Keywords— Precision Agriculture, IoT Sensors, Machine Learning, Soil Nutrient Monitoring, Crop Recommendation Systems.

I. INTRODUCTION

The agricultural sector is the most significant and is experiencing rapid transition, driven by technological advancements aimed at enhancing productivity and sustainability. The increasing global population demands the advancement of agricultural practices that are efficient and sustainable. Traditional methods of monitoring soil nutrient levels and controlling crops often inadequately address the complex and dynamic needs of modern agriculture. To address this discrepancy, the advancements of IoT and ML shows a significant method. One of the notable breakthroughs is the integration of IoT with ML technology, which is considered as a groundbreaking approach. The IoT facilitates the gathering of data in real-time by using a network of sensors that observe different soil characteristics, such as pH levels, moisture levels, temperature, and nutrient concentrations. Continuous monitoring enables farmers to promptly get insights into soil health, allowing for quick and accurate actions.

Machine learning enhances IoT by providing robust tools for data analysis and predictive modelling. ML

algorithms can analyse the extensive data produced by IoT sensors to recognise trends, identify deviations, and provide precise forecasts on soil nutrient deficits and the most suitable crop selections. Machine learning algorithms can analyse substantial amounts of data generated by IoT devices, uncovering hidden patterns and trends that may not be readily apparent.

By using predictive analytics and pattern recognition, ML provides a practical and useful information, such as the most effective schedules for fertilisation, methods for crop rotation, and early alerts for possible soil deficiencies. The integration of IoT and ML improves the accuracy of soil nutrient management and facilitates the creation of customised crop recommendations. The integration of IoT with ML has the capacity to transform soil nutrient management and crop recommendation systems, resulting in enhanced decision-making, optimised resource utilisation, and refined agricultural production.

The aim of the presented survey paper is to provide a thorough and comprehensive survey of the present status of research and development in this particular sector. This paper analyses the structure and operation of integrated IoT and ML systems. It explores the challenges and potential future scope in the field of precision agriculture.

Rest of the paper is organised as follows: Section II discusses about the background and related work to IoT and ML in agriculture sector. Section III presents the identified research gaps. Section IV describes the future direction for further investigation in precision agriculture by highlighting the scope of integrated IoT and ML technologies together. Section V provides the conclusion with key findings from presented literature survey.

II. LITERATURE REVIEW

Recent years have seen a lot of research on IoT and ML algorithms integration use for soil nutrient monitoring and crop suggestions generation. Emphasising significant contributions, methods, and discoveries, this literature review looks at major development work in this diverse research area.

A. IoT-Based Soil Monitoring Systems

The data provided by IoT-based devices is real-time in nature which has helped soil monitoring devices to acquire popularity. Advanced sensor networks built by researchers including [1][2] which assesses the temperature, soil moisture, pH, and nutrient levels. Their systems broadcast data to centralised platforms using wireless sensor networks (WSNs) and associated communication protocols.

In [3] authors continuously measured variables at the agriculture field. The implemented work has been validated on the smartphone platform with IoT, DSS and Real-time data. Authors in [4] implemented the automated remote field monitoring system and provides the visualization in the cloud with LoRaWAN and Real-time data. Authors in [5] introduced a low-cost and ecological IoT sensor system for precision farming, which significantly improved the accuracy of soil condition assessments.

B. Data Analysis with Machine Learning

ML techniques has been useful to analyse the vast amounts of data generated by IoT sensors. Studies in [6] proposes the Soil Classification based on micronutrients. Soil classification result in accuracy of 94 % with ELM and Private dataset. In [7] explored the crop recommendation for predictive accuracy of soil health and crop yield forecasts. Their work highlighted the potential of ML to uncover complex relationships between soil parameters and crop performance. Crop recommendation were predicted in accuracy of 98.22 % with MLP and Kaggle dataset. Authors in [8] implemented the system to aid the farmer in developing a platform that provides recommendations for crops. Results shows that crop recommendation by Random forest achieved 97.18 % accuracy with RF and Kaggle dataset.

C. Integration Challenges and Solutions

Integrating IoT and ML in agriculture involves several challenges, including data management, system interoperability, and scalability. Authors in [9], proposed the solution to provide consistent and up-to-date agricultural data and field updates to assist farmers. The model proposed by authors has achieved the 97.3 % of accuracy with MSVMDAG-FFO and own dataset. In [10], discussed the implementation using RFC and one's own dataset, crop recommendation achieved 99% accuracy. Also discussed the novel approaches to improve data accuracy and consistency. In [11], worked on crop selection with SCS and dataset from Pakistan achieved accuracy 97.4 %. Along with that addressed issues of system integration and interoperability, recommending standardized protocols and frameworks for seamless

communication between IoT devices and ML models. In [12], authors has implemented the system with the help of Amazon Web Service, arduino, and set of sensors. To analyse and visualise data, they made use of mobile applications, cloud computing, and machine learning techniques. This system makes it possible to ascertain the minerals that are present in the soil and prescribe the appropriate amount of fertiliser. Crop recommendation and soil nutrient monitoring are made more comprehensive by the integrating IoT and ML technology[13-21].

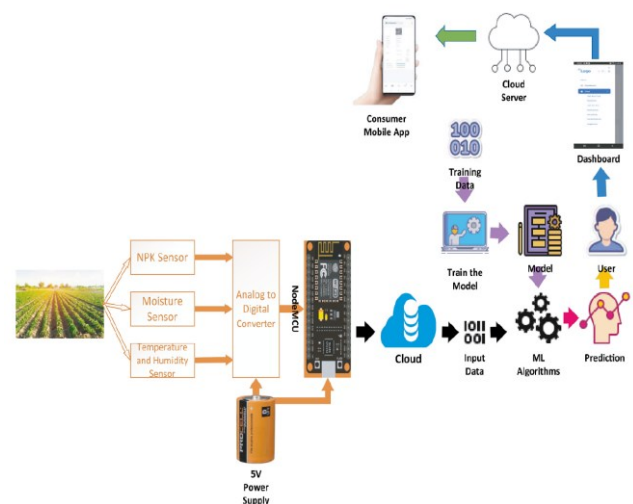


Fig. 1: Methodology framework for soil nutrients monitoring and crop recommendation using an advanced ML-Enabled IoT system [1].

The Figure 1 shows a framework having integration of ML and IoT technology. Suggested framework in [1] uses 3 sensors namely JXBS-3001, FC-28, and DHT11 to measure moisture, humidity, soil nutrient concentrations (NPK), and temperature in the agricultural field. The NodeMCU sends gathered data to a remote server via MQTT for processing. The mentioned sensors give updated data to the suggested machine learning algorithm for using data to recommend planting schedules, crop kinds, and fertilisation strategies. This helps farmers choose crops and boost output. In addition, soil monitoring assures the quality of food and implanted sensors analyse food dietary information and hazardous pollutants if available. The framework's User-Friendly Mobile App lets farmers remotely monitor and manage their agricultural activities with seamless device and data access.

Table 1 shows a brief review provided on the related work to monitor the real time soil nutrients and recommending the crops with the Machine learning enabled internet of things system.

TABLE 1: SUMMARY OF EXISTING LITERATURE SURVEY

Sr. No.	Authors	Year	Problem Statement	ML-IoT Enabled	Solution
1	Md Reazul Islam et al [1]	2023	The integration of ML models and IoT for monitor soil nutrients and recommend crops.	Yes	Innovative ML-enabled monitoring and recommendation system with IoT device and Real-time data.
2	S.J. Ramson et al [2]	2021	Soil health monitoring in UpToDate manner.	No	Development and evaluation of an economical monitoring system (5667 USD) with IoT and Real-time data.
3	S.V. Gaikwad et al [3]	2021	Measuring the agriculture parameters.	No	Develop and test the smartphones platform with IoT, DSS and Real-time data.
4	B.M. Zerihun et al [4]	2022	A system that monitors fields remotely using automation.	No	Visualization in the cloud with LoRaWAN and Real-time data.
5	V. Shukla et al [5]	2022	Soil monitoring system.	No	Identify soil type effectively and use IoT and real-time data to visually represent related information.
6	T. Blesslin Sheeba et al [6]	2022	Soil Classification based on micronutrients.	Yes	Soil classification result in accuracy of 94 % with ELM and Private dataset.
7	K. Bakthavatchalam et al [7]	2022	Crop recommendation.	Yes	Crop recommendation were predicted in accuracy of 98.22 % with MLP and Kaggle dataset.
8	Thilakarathne NN et al [8]	2022	To create a platform for crop recommendations in order to help farmers.	No	Crop recommendation by Random forest achieved 97.18 % accuracy with RF and Kaggle dataset.
9	Senapaty MK et al [9]	2023	To regularly provide farmers with up-to-date crop and field information.	Yes	The performance accuracy of 97.3 % with MSVMDAG-FFO and own dataset.
10	S. Sundaresan et al [10]	2023	Crop Recommendation system.	Yes	Using RFC and one's own dataset, crop recommendation achieved 99% accuracy.

11	A. Ikram et al [11]	2022	Correct Selection of Crop.	Yes	Crop selection with SCS and dataset from Pakistan achieved accuracy 97.4 %
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III. RESEARCH GAPS

After extensive literature survey the following gaps are identified in soil nutrients monitoring and crop recommendation:

- *Labor-intensive and subjective soil nutrient analysis:*

Conventional techniques for analysing soil nutrients and recommending crops frequently depend on labour-intensive, subjective evaluations.

- *Inefficient resource utilization:*

Ineffective resource management and fertiliser application choices can result in lower yields and inefficient use of resources.

- *Lack of live monitoring and adaptive decisions:*

Traditional agricultural practices often lack the real-time monitoring and adaptive decision making capabilities.

- *Limited access to crop quality assessment:*

Information about the quality of crops consumed by individuals is often not easily accessible.

- *Fragmented solutions and high costs:*

High costs and fragmented solutions are common problems with current IoT-based soil monitoring systems.

IV. FUTURE DIRECTION

- **Enhanced Sensor Technology:** The development of more advanced, reasonably affordable priced sensors with greater accuracy and longer battery life would improve real-time data collection and reduce maintenance requirements.
- **Data Integration and Management:** The accuracy and opportunity of the data collected will be increased by improving methods for combining data from different IoT devices and ensuring that data management systems are robust.
- **Advanced Machine Learning Algorithms:** Research on more enhanced ML algorithms, such as deep learning and reinforcement learning, may improve prediction accuracy and provide more insightful data for soil management and crop recommendations.
- **Edge Computing:** Integrating edge computing into local data processing on IoT devices may enhance real-time analytics, reduce latency, and boost system responsiveness, particularly in remote agricultural regions.

- **Interoperability Standards:** Standardized frameworks and protocols for ML models and IoT devices will enable smooth communication and integration between various platforms and systems.
- **Explainability and Transparency:** By making outcomes easier to understand and use, improving the interpretability of ML models can assist farmers in making better decisions.
- **Scalability and Cost-Effectiveness:** Research on scalable and cost-effective solutions will enable small and medium-sized farms to afford more advanced IoT-ML systems, hence promoting broader adoption.
- **Integration with Recent Technologies:** Research should be done on emerging technologies like blockchain for data security and augmented reality (AR) for visualisation in order to further enhance the capabilities and applications of IoT-ML systems in agriculture.

V. CONCLUSION

The integration of IoT and Machine Learning represents a significant development in soil nutrient monitoring and crop recommendation. The integration of real-time data collecting with IoT technology and advanced analytics using machine learning significantly improves precision agriculture, particularly in soil monitoring and crop production. Despite the ongoing hurdles of data integration, system interoperability, and algorithmic complexity, ongoing research and technology progress provide optimism for overcoming these challenges. Developments in sensor technology, data management, and machine learning algorithms will improve agricultural practices, development sustainability and efficiency in precision farming.

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